

REST API Reference Documentation

Overview

Hostel Outing Permission & Approval Management System This document provides the complete, authoritative API reference for the FastAPI backend service.

Base URL: `http://127.0.0.1:8000/api`

Endpoint Summary Table

Method	Endpoint	Allowed Role	Purpose
GET	<code>/api/health</code>	Public	System health check
POST	<code>/api/auth/signup</code>	Public	Register new Student, HOD, or Warden account
POST	<code>/api/auth/login</code>	Public	Authenticate user & retrieve JWT access token
GET	<code>/api/auth/me</code>	Authenticated	Retrieve profile of authenticated user
GET	<code>/api/departments</code>	Public	List all academic departments
GET	<code>/api/hostel-blocks</code>	Public	List all hostel blocks
POST	<code>/api/outings</code>	STUDENT	Submit a new outing permission request
GET	<code>/api/outings/my</code>	STUDENT	List all outing requests submitted by logged-in student
GET	<code>/api/outings/{id}</code>	Scoped Roles	Retrieve specific outing details
POST	<code>/api/outings/{id}/cancel</code>	STUDENT	Cancel eligible pending outing request
GET	<code>/api/outings/{id}/history</code>	Scoped Roles	Fetch complete audit timeline for an outing request
GET	<code>/api/hod/outings/pending</code>	HOD	List pending requests for HOD's department (PENDING_HOD)
GET	<code>/api/hod/history</code>	HOD	Query department outing history with search & filters
POST	<code>/api/hod/outings/{id}/approve</code>	HOD	Approve student request (advance to PENDING_WARDEN)

Method	Endpoint	Allowed Role	Purpose
POST	/api/hod/outings/{id}/reject	HOD	Reject student request (set to REJECTED)
GET	/api/warden/outings/pending	WARDEN	List pending requests for Warden's block (PENDING_WARDEN)
GET	/api/warden/history	WARDEN	Query hostel block outing history with search & filters
POST	/api/warden/outings/{id}/parent-confirmation	WARDEN	Confirm parent/guardian consent via phone call
POST	/api/warden/outings/{id}/approve	WARDEN	Grant final Warden approval (set to APPROVED)
POST	/api/warden/outings/{id}/reject	WARDEN	Reject student request at Warden stage
GET	/api/watchman/outings/today	WATCHMAN	List today's active gate outings
GET	/api/watchman/outings/search	WATCHMAN	Search gate outings by request ID, reg number, or name
GET	/api/watchman/students	WATCHMAN	Campus-wide student directory lookup
POST	/api/watchman/outings/{id}/exit	WATCHMAN	Record student gate exit (set to EXITED)
POST	/api/watchman/outings/{id}/return	WATCHMAN	Record student gate return (COMPLETED or LATE_RETURN)

Detailed Endpoint Specifications

1. System & Authentication Endpoints

1.1 GET /api/health

- **Method:** GET
- **Authentication:** None
- **Purpose:** Health check endpoint.
- **Response** (200 OK):

```
{
  "status": "healthy",
  "service": "Hostel Outing Management API"
}
```

1.2 POST /api/auth/signup

- **Method:** POST
- **Authentication:** None (Public)
- **Purpose:** Account registration for Students, HODs, and Wardens.
- **Request Body** (SignupRequest):

```
{
  "name": "Arjun Raj",
  "email": "student.a@hostelapp.local",
  "password": "Hostel@123",
  "confirm_password": "Hostel@123",
  "role": "STUDENT",
  "register_number": "CSE2027001",
  "department_id": 1,
  "hostel_block_id": 1,
  "year": 3,
  "room_number": "A-101"
}
```

- **Response** (201 Created - UserResponse):

```
{
  "id": 1,
  "name": "Arjun Raj",
  "register_number": "CSE2027001",
  "email": "student.a@hostelapp.local",
  "role": "STUDENT",
  "department_id": 1,
  "hostel_block_id": 1,
  "year": 3,
  "hostel": "A Block",
  "room_number": "A-101",
  "is_active": true
}
```

- **Possible Errors:**
 - 400 Bad Request: Password mismatch, invalid role, Watchman public signup attempt, duplicate email/register number, missing department/block ID, active HOD/Warden already assigned.

1.3 POST /api/auth/login

- **Method:** POST
- **Authentication:** None (Public)
- **Purpose:** Authenticate credentials and return JWT bearer token.

- **Request Body** (LoginRequest):

```
{
  "email": "student.a@hostelapp.local",
  "password": "Hostel@123"
}
```

- **Response** (200 OK - Token):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "user_id": 1,
  "name": "Arjun Raj",
  "email": "student.a@hostelapp.local",
  "role": "STUDENT",
  "register_number": "CSE2027001",
  "department_id": 1,
  "department_code": "CSE",
  "department_name": "Computer Science and Engineering",
  "hostel_block_id": 1,
  "hostel_block_name": "A Block",
  "year": 3
}
```

- **Possible Errors:**

- 401 Unauthorized: Invalid email or password.

1.4 GET /api/auth/me

- **Method:** GET
- **Authentication:** Bearer Token
- **Purpose:** Get current user profile.
- **Response** (200 OK - UserResponse).

1.5 GET /api/departments

- **Method:** GET
- **Authentication:** None
- **Purpose:** List all academic departments.
- **Response** (200 OK - List of DepartmentResponse):

```
[
  { "id": 1, "name": "Computer Science and Engineering", "code": "CSE" },
  { "id": 2, "name": "Electronics and Communication Engineering", "code": "ECE" }
]
```

1.6 GET /api/hostel-blocks

- **Method:** GET
- **Authentication:** None
- **Purpose:** List all hostel blocks.
- **Response** (200 OK - List of HostelBlockResponse):

```
[
  { "id": 1, "name": "A Block" },
  { "id": 2, "name": "B Block" },
  { "id": 3, "name": "C Block" }
]
```

2. Student Outing Endpoints

2.1 POST /api/outings

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** STUDENT
- **Purpose:** Submit a new outing permission request.
- **Request Body** (OutingCreate):

```
{
  "outing_date": "2026-08-15",
  "leaving_time": "09:00:00",
  "expected_return_time": "17:00:00",
  "destination": "City Library",
  "reason": "Research reference books."
}
```

- **Response** (201 Created - OutingResponse):

```
{
  "id": 4,
  "student_id": 1,
  "outing_date": "2026-08-15",
  "leaving_time": "09:00:00",
  "expected_return_time": "17:00:00",
  "destination": "City Library",
  "reason": "Research reference books.",
  "status": "PENDING_HOD",
  "parent_approval_confirmed": false,
  "created_at": "2026-08-12T20:30:00",
  "updated_at": "2026-08-12T20:30:00"
}
```

- **Possible Errors:**

- 400 Bad Request: Outing date in past, leaving time after return time, active overlapping request exists.

2.2 GET /api/outings/my

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** STUDENT
- **Purpose:** List outing requests created by the logged-in student.
- **Response** (200 OK - List of OutingResponse).

2.3 `GET /api/outings/

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** Scoped Roles (STUDENT can only access own outing; HOD/Warden/Watchman access scoped outings)
- **Path Parameter:** id (int) - Outing Request ID
- **Response** (200 OK - OutingResponse).
- **Possible Errors:**
 - 403 Forbidden: Unauthorized cross-student or cross-department access attempt.
 - 404 Not Found: Outing request does not exist.

2.4 `POST /api/outings/

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** STUDENT
- **Path Parameter:** id (int)
- **Purpose:** Cancel a pending outing request (PENDING_HOD or PENDING_WARDEN).

- **Response** (200 OK - OutingResponse with status: "CANCELLED").
- **Possible Errors:**
 - 400 Bad Request: Request is already approved, rejected, exited, or completed.
 - 403 Forbidden: Attempting to cancel another student's outing.

2.5 `GET /api/outings/`

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** Scoped Roles
- **Path Parameter:** id (int)
- **Purpose:** Fetch chronological audit timeline for an outing request.
- **Response** (200 OK - List of ApprovalHistoryResponse):

```
[
  {
    "id": 1,
    "outing_id": 4,
    "actor_id": 1,
    "actor_name": "Arjun Raj",
    "actor_role": "STUDENT",
    "action": "SUBMITTED",
    "comment": "Outing request submitted by student.",
    "timestamp": "2026-08-12T20:30:00"
  }
]
```

3. HOD Endpoints

3.1 GET /api/hod/outings/pending

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** HOD
- **Authorization Scope:** Department-scoped (User.department_id == current_user.department_id)
- **Purpose:** List pending requests (PENDING_HOD) for HOD's department.
- **Response** (200 OK - List of OutingResponse).

3.2 GET /api/hod/history

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** HOD

- **Query Parameters:**
 - `search` (string, optional): Search by student name, register number, or outing ID #OUT-X.
 - `status_filter` (string, optional): Filter by `OutingStatus`.
 - `hostel_block_id` (int, optional): Filter by hostel block.
- **Response** (200 OK - List of `OutingResponse`).

3.3 `POST /api/hod/outings/`

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** HOD
- **Path Parameter:** `id` (int)
- **Request Body** (`DecisionRequest`, optional):

```
{ "comment": "Academic permission granted." }
```

- **Response** (200 OK - `OutingResponse` with `status: "PENDING_WARDEN"`).
- **Possible Errors:**
 - 400 Bad Request: Outing not in `PENDING_HOD` status.
 - 403 Forbidden: Request belongs to a student outside HOD's department.

3.4 `POST /api/hod/outings/`

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** HOD
- **Path Parameter:** `id` (int)
- **Request Body** (`DecisionRequest`, optional):

```
{ "comment": "Academic review rejected due to upcoming midterms." }
```

- **Response** (200 OK - `OutingResponse` with `status: "REJECTED"`).

4. Warden Endpoints

4.1 GET /api/warden/outings/pending

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** WARDEN
- **Authorization Scope:** `Hostel-block-scoped (User.hostel_block_id == current_user.hostel_block_id)`
- **Purpose:** List pending requests (`PENDING_WARDEN`) for Warden's hostel block.

- **Response** (200 OK - List of OutingResponse).

4.2 GET /api/warden/history

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** WARDEN
- **Query Parameters:**
 - `search` (string, optional)
 - `status_filter` (string, optional)
 - `department_id` (int, optional)
- **Response** (200 OK - List of OutingResponse).

4.3 `POST /api/warden/outings/

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** WARDEN
- **Path Parameter:** `id` (int)
- **Purpose:** Mark parent/guardian consent confirmed via phone call.
- **Response** (200 OK - OutingResponse with `parent_approval_confirmed: true`).
- **Possible Errors:**
 - 400 Bad Request: Outing not in pending warden status.
 - 403 Forbidden: Request belongs to a student outside Warden's hostel block.

4.4 `POST /api/warden/outings/

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** WARDEN
- **Path Parameter:** `id` (int)
- **Request Body** (DecisionRequest, optional):

```
{ "comment": "Parent phone confirmation received; final approval granted." }
```

- **Response** (200 OK - OutingResponse with `status: "APPROVED"`).
- **Possible Errors:**
 - 400 Bad Request: Parent approval has not been confirmed (`parent_approval_confirmed == False`).

4.5 `POST /api/warden/outings/

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** WARDEN

- **Path Parameter:** id (int)
- **Request Body** (DecisionRequest, optional):

```
{ "comment": "Parent declined permission." }
```

- **Response** (200 OK - OutingResponse with status: "REJECTED").
-

5. Watchman Endpoints

5.1 GET /api/watchman/outings/today

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** WATCHMAN
- **Purpose:** Retrieve active gate outings for today (APPROVED, EXITED, COMPLETED, LATE_RETURN).
- **Response** (200 OK - List of OutingResponse).

5.2 GET /api/watchman/outings/search

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** WATCHMAN
- **Query Parameter:** query (string, required) - Search string
- **Response** (200 OK - List of OutingResponse).

5.3 GET /api/watchman/students

- **Method:** GET
- **Authentication:** Bearer Token
- **Allowed Role:** WATCHMAN
- **Query Parameters:**
 - search (string, optional)
 - department_id (int, optional)
 - hostel_block_id (int, optional)
- **Response** (200 OK - List of StudentDirectoryResponse).

5.4 `POST /api/watchman/outings/

- **Method:** POST
- **Authentication:** Bearer Token
- **Allowed Role:** WATCHMAN
- **Path Parameter:** id (int)
- **Purpose:** Record student exit at main gate.
- **Response** (200 OK - OutingResponse with status: "EXITED").

- **Possible Errors:**

- 400 Bad Request: Request status is not APPROVED or exit was already recorded.

5.5 `POST /api/watchman/outings/`

- **Method:** POST

- **Authentication:** Bearer Token

- **Allowed Role:** WATCHMAN

- **Path Parameter:** id (int)

- **Purpose:** Record student return at main gate and evaluate punctuality.

- **Response** (200 OK - OutingResponse):

- **On-Time:** status: "COMPLETED"
- **Late Return:** status: "LATE_RETURN" with delay minutes logged in gate_log and LATE_RETURN_DETECTED audit record.

- **Possible Errors:**

- 400 Bad Request: Return recorded before exit, or return already recorded.